

Problem 1. Two blocks are connected by a massless rope, one on a table and the other suspended from a pulley to hang below the table. The mass of the block on the table is 10 kg and the hanging mass is 15 kg. The table is frictionless, and the pulley is frictionless and massless. (a) Draw the situation and find the acceleration of the blocks. (b) Find the tension in the rope. (c) Find the speed with which the hanging mass hits the floor if it starts from rest and is initially located .3 m from the floor.

Solution.

(a) Find the acceleration

Step I: Write Newton's 2nd law for each block. For the block on the table:

$$\sum F_x = m_1 a \implies T = m_1 a \quad (1)$$

For the hanging block, let downward be positive:

$$\begin{aligned} \sum F_y &= m_2 a \\ m_2 g - T &= m_2 a \end{aligned} \quad (2)$$

Step II: Solve the system. By plugging in the result from equation into the hanging-mass equation:

$$\begin{aligned} m_2 g - m_1 a &= m_2 a \\ m_2 g &= m_2 a + m_1 a \\ m_2 g &= (m_2 + m_1) a \\ \implies a &= \frac{m_2 g}{(m_2 + m_1)} \end{aligned}$$

Plug in the numbers:

$$a = \frac{(15 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})}{(15 \text{ kg} + 10 \text{ kg})} = \frac{147}{25} \frac{\text{m}}{\text{s}^2} = 5.9 \frac{\text{m}}{\text{s}^2}$$

(b) Find the tension, use equation (1).

$$\begin{aligned} T &= m_1 a \\ &= (10 \text{ kg})(5.9 \frac{\text{m}}{\text{s}^2}) = 59 \text{ N} \end{aligned}$$

You can also check with the other block if you don't believe me... use equation (2).

$$\begin{aligned} m_2 g - T &= m_2 a \\ (15 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2}) - T &= (15 \text{ kg})(5.9 \frac{\text{m}}{\text{s}^2}) \\ T &= (15 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2}) - (15 \text{ kg})(5.9 \frac{\text{m}}{\text{s}^2}) = 147 \text{ N} - 88.5 \text{ N} = 58.5 \text{ N} \end{aligned}$$

- (c) Find the speed when the hanging mass hits the floor. Since the hanging mass starts with from rest and falls 3m. Use kinematics:

$$v^2 = v_0^2 + 2a\Delta y \quad (3)$$

Since it starts from rest $v_0 = 0$, so

$$\begin{aligned} v^2 &= 2a\Delta y \\ &= 2(5.9 \frac{\text{m}}{\text{s}^2})(.3 \text{ m}) \\ &= 3.54 \frac{\text{m}^2}{\text{s}^2} \\ v &= \sqrt{3.54} \frac{\text{m}}{\text{s}} = 1.88 \frac{\text{m}}{\text{s}} \end{aligned}$$

Problem 2. In the simple situation where something travels in a circular orbit around another object. Let the radius of the orbit be 6 m, and the speed of the object moving be 4 m/s. The mass of the object be 10 kg. What is the force on the electron?

Solution.

Use centripital force equation

$$F_c = \frac{mv^2}{r} = \frac{10\text{kg}(4\text{m/s})^2}{6 \text{ m}} = \frac{160}{6} = 26.6 \text{ N} \quad (4)$$